Antimatter at CERN: Precision Symmetry Tests from Antiprotons to Antihydrogen

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Abstract

This review examines CERN's low-energy antimatter program as a coordinated precision-measurement ecosystem rather than a single search for missing cosmic antimatter. Using CERN's AD/ELENA infrastructure as the organizing center, it compares how BASE, ALPHA, AEgIS, GBAR, ASACUSA, and PUMA turn broad questions about matter-antimatter symmetry, gravity, and nuclear structure into experimentally distinct observables. BASE constrains baryonic symmetry through Penning-trap measurements of antiprotons and protons; ALPHA develops antihydrogen as a platform for spectroscopy, accumulation, cooling, and gravity studies; AEgIS and GBAR pursue independent antihydrogen gravity strategies; ASACUSA uses antiprotonic atoms and antihydrogen spectroscopy; and PUMA applies antiprotons to rare nuclear systems. Recent 2023-2026 results show the field moving from first demonstrations toward improved control, including direct antihydrogen gravity observation, coherent single-antiproton spin spectroscopy, larger antihydrogen samples, and ppm-scale hyperfine measurements. The review argues that the program's scientific value lies in complementarity: different experiments test different observables and systematic limits, together clarifying what current antimatter measurements establish and what remains open.

1. Introduction

Antimatter research is often introduced through one large question: why does the observable universe contain so much matter and so little antimatter? CERN's public overview frames the puzzle in this way, noting that particle-antiparticle pairs should have been abundant in the early universe, yet the visible universe today is overwhelmingly matter [1]. That question is real, and it gives antimatter experiments their broad significance. But it can also obscure what laboratory antimatter experiments actually measure. A trapped antiproton, an antihydrogen atom, or an antiprotonic helium transition does not directly replay baryogenesis. It provides a controlled comparison between matter and antimatter in one observable at a time.

This distinction is the starting point for the present review. Baryogenesis is a theory problem about how a matter excess could have emerged dynamically in the early universe, with requirements such as baryon-number violation, C and CP violation, and departure from thermal equilibrium [2]. CERN's low-energy antimatter program addresses a related but narrower experimental question: do matter and antimatter behave identically in the properties and interactions that can be measured now? The connection is important, but indirect. If a precise matter-antimatter difference were found, it could point toward physics beyond the current framework. If no difference is found, the allowed space for such physics becomes more constrained.

The central claim of this paper is that CERN's antimatter program is best understood as a coordinated precision-measurement ecosystem rather than as a single search for missing antimatter. Its experiments do not all ask the same question. BASE compares protons and antiprotons in Penning traps. ALPHA studies trapped antihydrogen through spectroscopy, cooling, accumulation, and gravity measurements. AEgIS and GBAR pursue independent strategies for antimatter gravity. ASACUSA uses antiprotonic helium and antihydrogen spectroscopy. PUMA uses antiprotons as probes of nuclear surfaces. The 2025 AD/ELENA program review captures this breadth by describing high-precision work on antiprotons, antihydrogen, and antiprotonic atoms across CPT tests, gravity, and nuclear-structure studies [3].

The review therefore has two aims. The first is explanatory: to make the major CERN antimatter experiments understandable by linking each one to its physical system, observable, and limitation. The second is interpretive: to argue that the program's value lies in complementarity. A charge-to-mass ratio comparison, a hyperfine splitting measurement, a gravitational release experiment, a beam deflectometer, and an antiproton-nucleus annihilation study should not be forced onto one scale of importance. They probe different assumptions and face different systematic risks.

The paper proceeds in that order. Section 2 defines the concepts and observables needed to read antimatter results carefully. Section 3 explains why AD/ELENA infrastructure is the enabling condition for the field. Section 4 maps the experiments by measurement role. Section 5 compares their observables and limitations directly. Section 6 reviews recent 2023-2026 results and the open questions they leave behind. The goal is not to claim a new experimental result, but to build a rigorous framework for understanding what current antimatter measurements do and do not show.

2. Concepts And Observables

Antimatter experiments are easiest to misunderstand when they are described only by their dramatic motivation. The missing cosmic antimatter is real as a problem, but CERN's low-energy antimatter program does not measure the universe's baryon asymmetry directly. It measures particular systems, such as antiprotons, antihydrogen, and antiprotonic atoms, under carefully controlled conditions. The concepts that matter for this review are therefore not only "matter versus antimatter," but also which property is being compared, at what precision, and under which experimental assumptions.

An antiparticle is not an exotic opposite of matter in every possible sense. In the standard description, each matter particle has a corresponding antiparticle with the same mass and opposite conserved charges. CERN's public overview uses this fact to introduce both the basic symmetry and the puzzle: if particle-antiparticle pairs were produced together in the early universe, then the observed matter-dominated universe requires an additional explanation [1]. That puzzle motivates antimatter research, but it should not be collapsed into a single experimental target. Baryogenesis reviews frame the problem as a set of dynamical conditions for generating a matter excess in the early universe, involving ingredients such as baryon-number violation, C and CP violation, and departure from thermal equilibrium [2]. Precision tests with antiprotons and antihydrogen can probe symmetry assumptions relevant to the bigger picture, but they do not by themselves reconstruct the early-universe mechanism.

The central symmetry language for this paper is CPT: charge conjugation, parity reversal, and time reversal considered together. In ordinary local relativistic quantum field theory, CPT symmetry implies that particles and antiparticles should have matched properties in specific conjugate comparisons, such as equal masses and opposite charges. Experimental CPT tests therefore become comparisons of observables, not declarations that "CPT is proven" in a universal sense. A charge-to-mass ratio, a magnetic moment, an atomic transition frequency, and a hyperfine interval are all different windows into the same broad question. Greenberg's result that CPT violation in an interacting theory implies Lorentz violation is useful here because it shows why CPT tests are tied to deeper spacetime-symmetry assumptions rather than being isolated bookkeeping exercises [4]. The practical rule for this paper is simple: each claim about symmetry must name the observable and the precision, not merely the principle.

For charged antiparticles, Penning traps turn those symmetry questions into frequency measurements. In an idealized magnetic field, the cyclotron frequency of a charged particle is related to its charge-to-mass ratio by

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omega_c = |q|B / m
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where `q` is charge, `B` is magnetic-field strength, and `m` is mass. Real Penning-trap experiments require a much richer treatment of trap motion, field imperfections, image currents, cooling, and systematic shifts; the classic Brown and Gabrielse review provides the theoretical language for single charged particles in such traps [5]. This is the conceptual basis for why BASE can test matter-antimatter symmetry without making antihydrogen. By comparing proton and antiproton properties in cryogenic Penning traps, BASE measures baryonic observables directly. Its 2022 charge-to-mass comparison reported a result consistent with CPT at a fractional uncertainty of 16 parts per trillion [6]. That result is not a general solution to the antimatter puzzle. It is a sharply defined test of one conjugate baryonic property.

Neutral anti-atoms open a different route. Antihydrogen, made from an antiproton and a positron, can be compared with hydrogen through atomic transitions. This is powerful because hydrogen is one of the most precisely studied systems in physics, but it is also difficult because antihydrogen must be produced, cooled, trapped, and detected through annihilation signatures. ALPHA's observation of the 1S-2S transition in trapped antihydrogen established a route toward hydrogen-antihydrogen spectroscopy [7]. Later, laser cooling narrowed spectral lines and improved control over antihydrogen motion [8]. More recent hyperfine-component spectroscopy and ground-state hyperfine measurements show how antihydrogen is becoming a more mature precision platform rather than only a proof-of-existence achievement [9, 10]. These measurements probe internal anti-atomic structure and symmetry by asking whether antihydrogen behaves like the CPT conjugate of hydrogen within the measured uncertainty.

Gravity introduces another observable family. The weak equivalence principle says that freely falling bodies follow the same trajectory in a gravitational field independent of composition and internal structure, at least when only gravity acts. For antimatter, the challenge is not just conceptual but experimental. Charged antiparticles are extraordinarily sensitive to electromagnetic fields, which can overwhelm gravity. Neutral antihydrogen is therefore a better test body, but it must be cold enough and controlled well enough for gravitational effects to be separated from magnetic and kinetic effects. The older antimatter-gravity literature is valuable because it shows that "antigravity" has long been debated through conservation laws, equivalence-principle reasoning, and particle-physics constraints, not just through intuition [11]. ALPHA's 2023 ALPHA-g result moved this discussion from indirect argument to direct experiment by observing antihydrogen behavior consistent with gravitational attraction and ruling out simple repulsive antigravity for that system [12].

A useful simplified intuition is ordinary vertical motion,

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Delta z = (1/2) g t^2
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but this equation should not be mistaken for a full model of trapped antihydrogen release. In ALPHA-g, magnetic field gradients, trap geometry, atom energy distributions, annihilation reconstruction, and release protocols all matter. AEgIS and GBAR pursue different strategies, including beam deflectometry and ultracold antihydrogen free-fall concepts, precisely because each method controls different systematic risks. This is why gravity experiments should be compared by their kinematics and systematics, not only by the common question "does antimatter fall?"

These concepts set up the main argument of the paper. CERN's antimatter program is not a single experiment chasing a single yes-or-no answer. It is a set of measurement strategies that translate large questions about symmetry, gravity, and cosmic history into observables that can actually be tested: trap frequencies, spin transitions, atomic spectra, annihilation positions, and nuclear signatures. The next section therefore turns from concepts to infrastructure, because none of these observables exists without a machine capable of producing, slowing, and delivering antiprotons at usable energies.

3. AD/ELENA As Infrastructure

The experiments reviewed in this paper depend on a common fact that is easy to hide behind the names of the collaborations: low-energy antimatter is not simply found, collected, and measured. It has to be manufactured as a beam of antiprotons, slowed to useful energies, cooled so that the beam becomes experimentally manageable, captured in traps, and then transformed into the particular system each experiment requires. The Antiproton Decelerator (AD) and ELENA are therefore not just background machinery. They are the infrastructure that makes CERN's low-energy antimatter program possible.

CERN's public description of the AD begins with a useful contrast: not all accelerators are built to increase particle energy [13]. Antiprotons are produced when a proton beam from the Proton Synchrotron strikes a metal target. Those collisions create many secondary particles, including antiprotons, but the antiprotons are initially too energetic and too widely spread in direction and energy to be useful for precision antimatter experiments. A high-energy collision product is not yet a precision object. Before it can be stored, combined with positrons, or compared with a proton, it must be turned into a low-energy beam with a small enough phase-space spread for traps and beamlines to accept it.

The AD performs the first major transformation. CERN describes it as a ring of bending and focusing magnets, with strong electric fields used to slow antiprotons and cooling cycles used to reduce their energy spread and orbit deviations [13]. In practical terms, this changes the role of antiprotons from collision debris into a beam that can be delivered to experiments. The distinction matters for the argument of this review because precision comparisons are limited not only by theoretical questions but also by the ability to hold antimatter long enough and quietly enough to measure it. A Penning-trap frequency comparison, an antihydrogen spectroscopy run, and an antiprotonic-atom measurement all begin with the same infrastructure problem: making antiprotons slow and controllable.

ELENA, the Extra Low ENergy Antiproton ring, extends that transformation. CERN states that ELENA is a 30 m synchrotron coupled to the AD that reduces antiproton energy by a factor of 50, from 5.3 MeV to 0.1 MeV, while electron cooling increases the beam density [13]. This is not a cosmetic improvement. Lower-energy antiprotons are easier for experiments to trap, and CERN reports that ELENA increases the number of antiprotons that can be trapped by a factor of 10 to 100 [13]. In the logic of the paper, that increase is a bridge between facility engineering and physics reach: more captured antiprotons can mean higher measurement rates, better statistics, more systematic checks, or entirely new experimental modes.

The AD/ELENA program also explains why the CERN antimatter effort should be read as a coordinated portfolio rather than a set of isolated experiments. The 2025 AD/ELENA program review frames the facility as supporting high-precision studies of antiprotons, antihydrogen, and antiprotonic atoms, including CPT tests, antimatter gravity, dark-sector questions, and antiproton-based nuclear studies [3]. The same antiproton source therefore feeds experiments that look very different at the detector level. BASE stores charged antiprotons and protons in Penning traps for frequency and spin measurements. ALPHA combines antiprotons with positrons to produce trapped antihydrogen. AEgIS and GBAR pursue gravity-sensitive antihydrogen strategies. ASACUSA studies antiprotonic helium and antihydrogen spectroscopy. PUMA uses antiprotons as probes of neutron-rich nuclei. The common infrastructure is what allows these different experimental languages to belong to one antimatter program.

This infrastructure also exposes a basic limit: antimatter precision is rate-limited and handling-limited. CERN emphasizes that antimatter must be isolated from ordinary matter because contact causes annihilation, and that the Antimatter Factory delivers finite numbers of antiprotons that experiments capture only with limited efficiency [1]. The numbers are tiny by ordinary material standards. CERN describes production in terms of hundreds of millions of antiprotons per hour when operating, yet also notes that the corresponding mass is fantastically small [1]. That mismatch between large particle counts and negligible macroscopic mass is central to the experimental culture of antimatter physics. These are not bulk samples; they are rare, fragile systems from which information has to be extracted particle by particle or atom by atom.

Recent antihydrogen work illustrates why improvements in production, cooling, and accumulation are not merely technical footnotes. In 2025, ALPHA reported an eight-fold increase in the antihydrogen trapping rate using sympathetic cooling of positrons with laser-cooled beryllium ions, accumulating more than 15000 antihydrogen atoms in under seven hours [14]. That result belongs in an infrastructure section because it changes what later physics measurements can attempt. Larger trapped samples can support more systematic studies, sidereal searches, spectroscopy campaigns, and gravity measurements. In other words, method development expands the domain of possible questions.

AD/ELENA therefore defines the practical boundary of CERN's antimatter program. The facility does not answer CPT, gravity, or baryogenesis questions by itself. Instead, it creates the conditions under which those questions can be translated into measurable observables: trap frequencies, spin states, spectral lines, annihilation vertices, beam deflections, and nuclear annihilation products. The next section can now treat the individual experiments not as separate curiosities, but as different uses of the same scarce and carefully prepared antiproton resource.

4. Experiment Map

The CERN antimatter program is clearer if the experiments are grouped by what they measure rather than by acronym. Each collaboration receives antiprotons from the same infrastructure, but the antiprotons are converted into different physical systems: trapped charged particles, neutral antihydrogen, antihydrogen beams, antiprotonic helium, or antiproton-nucleus annihilation events. The result is not one linear experiment but a portfolio of measurement strategies.

Experiment	System	Main observable family	Role in this review
ALPHA	Trapped antihydrogen	Spectroscopy, gravity, accumulation, cooling	Main neutral anti-atom precision platform.
BASE	Single antiprotons/protons in Penning traps	Charge-to-mass ratio, magnetic moment, spin control	Main charged-baryon precision platform.
AEgIS	Antihydrogen beam strategy	Moire deflectometry / gravity	Gravity method contrast using beam geometry.
GBAR	Ultracold antihydrogen strategy	Free-fall gravity	Gravity method contrast using ultracold anti-atoms.
ASACUSA	Antiprotonic helium and antihydrogen	Spectroscopy, mass-ratio tests, collision studies	Exotic-atom precision role.
PUMA	Low-energy antiprotons with rare isotopes	Annihilation products / neutron-proton density tails	Nuclear-structure expansion of antimatter program.

ALPHA is the central antihydrogen platform in the program. CERN describes ALPHA as an experiment that makes, captures, and studies antihydrogen atoms in order to compare them with hydrogen [15]. The crucial shift from earlier antihydrogen production was confinement. If antihydrogen atoms reach ordinary matter walls, they annihilate quickly; if they can be magnetically trapped, they can become subjects for spectroscopy and gravity studies. CERN notes that ALPHA reported trapping antihydrogen for more than 16 minutes in 2011, long enough to begin detailed measurements [15]. That achievement is important historically because it turned antihydrogen from a produced anti-atom into an experimentally interrogable system.

The later ALPHA record shows why this matters. The 1S-2S transition observation established antihydrogen spectroscopy as a direct hydrogen-antihydrogen comparison [7]. Laser cooling then improved control over the motion of trapped antihydrogen and narrowed spectroscopy lines [8]. The 2023 ALPHA-g gravity result used a vertically oriented trap to release antihydrogen and reconstruct where annihilations occurred, finding behavior consistent with gravitational attraction rather than simple repulsive antigravity [12]. More recent work on hyperfine spectroscopy and accumulation shows the same program becoming more mature: faster line characterization, larger samples, and ppm-scale ground-state hyperfine measurements are steps toward systematic precision rather than isolated milestones [9, 14, 10].

BASE approaches matter-antimatter comparison differently. Instead of creating neutral anti-atoms, it stores charged particles in Penning traps and compares protons with antiprotons. CERN describes BASE as aiming for highly precise measurements of magnetic moments and charge-to-mass ratios, using a multi-Penning-trap system in which different traps identify spin state, flip spin, cool particles, and store antiprotons [16]. This makes BASE the cleanest example of baryonic single-particle precision in the CERN antimatter program. Its 2022 charge-to-mass result reported proton-antiproton agreement within 16 parts per trillion [6]. Its 2025 coherent single-antiproton spin spectroscopy result then shifted the emphasis toward quantum control: observing Rabi oscillations and long spin coherence offers a path to sharper magnetic-moment comparisons [17].

The contrast between ALPHA and BASE is the first major example of complementarity. ALPHA asks whether antihydrogen, as a neutral anti-atom, matches hydrogen internally and gravitationally. BASE asks whether antiprotons, as individual charged baryons, match protons in charge-to-mass ratio and magnetic behavior. The two programs do not duplicate each other. They test different systems, face different systematic effects, and translate symmetry into different observables. Agreement in one does not remove the need for the other.

AEgIS and GBAR further show why antimatter gravity cannot be reduced to one method. CERN describes AEgIS as aiming to measure Earth's gravitational acceleration on antihydrogen using a beam of antihydrogen atoms passed through a Moire deflectometer and a position-sensitive detector [18]. The principle is geometrical: a pattern formed by gratings is shifted by acceleration during flight, so the vertical displacement and time of flight can be used to infer gravity. A 2014 AEgIS-related Nature Communications paper demonstrated a moire deflectometer for antimatter using slow antiprotons and argued that the technique is applicable to antihydrogen measurements [19]. AEgIS is therefore best understood as a beam-and-deflectometry strategy.

GBAR takes a colder route. CERN describes GBAR as forming antihydrogen ions, cooling them to microkelvin temperatures, stripping the extra positron, and allowing neutral antihydrogen to fall from a height of 20 cm before annihilation detection [20]. The conceptual appeal is that ultracold antihydrogen should reduce the initial-motion uncertainty that makes gravity measurements difficult. The Walz and Haensch proposal for measuring antimatter gravity with ultracold antihydrogen gives this strategy its underlying logic [21]. Together with ALPHA's trapped-release method, AEgIS and GBAR show why the field benefits from independent gravitational strategies with different kinematics and systematic controls.

ASACUSA broadens the program beyond antihydrogen alone. CERN describes ASACUSA as comparing matter and antimatter using antiprotonic helium and antihydrogen, while also studying matter-antimatter collisions [22]. Antiprotonic helium is a hybrid exotic atom in which an antiproton replaces an electron in an atomic system. This makes it a precise testbed for spectroscopy and mass-ratio comparisons. ASACUSA's antiprotonic-helium work, including the 2016 buffer-gas cooling result and antiproton-to-electron mass-ratio determination, shows that antiprotons can probe symmetry through exotic atomic systems, not only through free antiprotons or antihydrogen [23].

PUMA is the most visibly different member of the group because it uses antimatter as a nuclear probe. CERN describes PUMA as a compact experiment designed to transport antiprotons from the Antimatter Factory to ISOLDE, where short-lived radioactive nuclei are produced [24]. In PUMA, antiprotons are stored and then mixed with unstable nuclei; the detector records annihilation products. The goal is to infer the relative densities of protons and neutrons at the nuclear surface, including features such as neutron skins and halos [24]. The PUMA collaboration's 2022 paper frames low-energy antiprotons as uniquely sensitive to the tail of nuclear density in stable and rare isotopes [25]. This moves antimatter research from symmetry comparison into nuclear-structure diagnostics.

The experiment map therefore supports the central thesis of this review. CERN's antimatter program is not one hierarchy in which every experiment asks the same question with different hardware. It is a deliberately varied ecosystem. BASE turns symmetry into trapped-particle frequencies and spin dynamics. ALPHA turns symmetry into anti-atomic spectra and trapped anti-atom motion. AEgIS and GBAR turn gravity into beam deflection or ultracold free fall. ASACUSA turns antiprotons into exotic atoms for spectroscopy. PUMA turns antiproton annihilation into information about nuclear density. The next section can compare these observables directly, because the central question is now not merely what each experiment does, but what each can measure better than the others.

5. Comparative Analysis

The experiment map shows that CERN's antimatter program is broad, but breadth alone is not the main point. The stronger interpretation is that each experiment makes a different kind of precision possible. BASE makes single charged baryons stable measurement objects. ALPHA makes neutral anti-atoms spectroscopic and gravitational test bodies. AEgIS and GBAR convert gravity into alternative trajectory problems. ASACUSA turns antiprotonic atoms into precision spectroscopy systems. PUMA uses annihilation not as a nuisance, but as a nuclear-structure signal. The experiments are therefore complementary because their observables, systematic limitations, and physics interpretations differ.

Observable	System	Experiment(s)	What it tests or probes	Main limitation to explain
Charge-to-mass ratio	Proton/antiproton	BASE	Baryonic CPT consistency for a specific conjugate observable	Trap fields, frequency precision, long-term systematics
Magnetic moment / spin	Single antiproton	BASE	Matter-antimatter symmetry in magnetic moments	Spin-state readout, coherence, magnetic-field stability
1S-2S / hyperfine spectroscopy	Antihydrogen	ALPHA	Hydrogen-antihydrogen comparison, SME-linked constraints, antiproton internal-structure sensitivity	Atom number, temperature, line width, magnetic-field control
Gravity response	Antihydrogen	ALPHA, AEgIS, GBAR	WEP and antimatter gravity	Initial kinetic energy, trajectories, magnetic gradients, statistics
Antiprotonic atom spectra	Antiprotonic helium	ASACUSA	Antiproton-to-electron mass ratio and exotic-atom QED comparisons	Cooling, line widths, QED theory precision
Annihilation signatures	Antiprotons on nuclei	PUMA	Nuclear density tails, neutron skins, halos	Antiproton transport, rare-isotope lifetimes, annihilation reconstruction

The most precise-looking numbers in the program come from experiments that can keep the measured system small and controlled. BASE's charge-to-mass comparison is the clearest example. Its 2022 result reported `-(q/m)_p/(q/m)_pbar = 1.000000000003(16)`, a 16 parts-per-trillion fractional uncertainty based on four long-term studies over 1.5 years [6]. The intellectual force of the result comes from narrowing the comparison: this is not a measurement of every possible difference between matter and antimatter, but a highly constrained comparison of one baryonic observable. That narrowness is a strength. It is what makes the result interpretable.

BASE's 2025 single-antiproton spin spectroscopy shows the same logic at the level of method development. The result is not simply another number in a table of CPT tests. It demonstrates coherent quantum transition spectroscopy of a single antiproton spin, including Rabi oscillations, spin-inversion probabilities above 80 percent, coherence times around 50 s, and resonance linewidths 16 times narrower than previous measurements [17]. Those details matter because magnetic-moment comparisons are limited by the ability to prepare, manipulate, and read out spin states in a stable magnetic environment. The measurement improves the experimental language available for future symmetry tests.

ALPHA's strength is almost the opposite. Its central object is not an isolated charged particle but antihydrogen, a neutral anti-atom whose internal structure and motion can be compared with hydrogen. This makes the measurement program richer but also more difficult. The anti-atoms must be produced from antiprotons and positrons, trapped magnetically, cooled, sometimes spin-prepared, and detected through annihilation. The payoff is that antihydrogen can test symmetry through atomic observables that charged antiprotons alone cannot provide. ALPHA's 2026 ground-state hyperfine result, for example, reports a 4 ppm measurement using roughly 24,000 anti-atoms, with `a_1S/h = 1,420,404.8 +/- 1.1(stat.) +/- 5.6(sys.) kHz` in a 1 T magnetic field [10]. That result is valuable not because it is the same kind of precision as BASE's charge-to-mass ratio, but because it probes a different system and a different sensitivity: antihydrogen hyperfine structure is linked to the internal structure of the antiproton.

Gravity measurements add another layer of difference. ALPHA's 2023 gravity result is often summarized as showing that antimatter falls down, but the paper's own claim is more careful: antihydrogen released from magnetic confinement behaved in a way consistent with gravitational attraction to Earth, and repulsive antigravity was ruled out in that tested case [12]. This caution is not a weakness. It is the right way to read a first direct result in a difficult system. The experiment does not close all questions about antimatter gravity; it establishes a direct experimental foothold and points toward future precision measurements of the magnitude of gravitational acceleration for anti-atoms.

AEgIS and GBAR matter because they make that foothold less method-dependent. AEgIS pursues a beam strategy in which antihydrogen passes through a Moire deflectometer so that gravitational acceleration can be inferred from a shifted pattern [18, 19]. GBAR pursues an ultracold strategy in which antihydrogen ions are cooled, neutralized, and allowed to fall from a known height [20, 21]. These strategies are not just alternative engineering preferences. They change the initial conditions, dominant systematics, and analysis geometry. Agreement among trapped-release, beam-deflection, and ultracold-free-fall approaches would be more persuasive than repetition of a single method.

ASACUSA and PUMA show that the program's complementarity extends beyond the usual pair of CPT and gravity. ASACUSA uses antiprotonic helium and antihydrogen spectroscopy to compare matter-antimatter properties in exotic atomic systems [22]. The 2016 antiprotonic-helium result, involving buffer-gas cooling and antiproton-to-electron mass-ratio determination, belongs to the same broad symmetry-testing ecosystem but uses a different atom and different theory inputs [23]. PUMA moves still further, using low-energy antiprotons to probe neutron and proton densities in rare nuclei through annihilation products [24, 25]. In that setting, annihilation is not just the event that destroys the antimatter sample. It is the measurement channel.

This comparison clarifies the paper's central claim. CERN's antimatter program should not be ranked by a single scale of importance or precision. A 16 ppt Penning-trap comparison, a 4 ppm hyperfine measurement, a first direct gravity observation, an antihydrogen beam strategy, and an antiprotonic nuclear probe do not compete in a simple ladder. They answer different questions under different constraints. The program's scientific value comes from the fact that these measurements can eventually cross-check one another: charged baryons against neutral anti-atoms, internal spectra against gravitational motion, and symmetry tests against nuclear probes. That is why the next question is not only what has been measured, but which recent results have changed the field's practical frontier.

6. Recent Results, Limits, And Future Directions

The most recent phase of CERN's antimatter program is best described as a shift from first demonstrations toward control. Earlier milestones established that antihydrogen could be produced, trapped, and spectroscopically interrogated. The 2023-2026 results suggest a more demanding stage: larger antihydrogen samples, better magnetic control, direct gravity measurements, coherent antiproton spin manipulation, and spectroscopy that begins to probe internal antiproton structure. The field is still limited by statistics and systematics, but the practical frontier has moved.

The 2023 ALPHA gravity result is the clearest public milestone in this period because it converted a long-standing question into a direct laboratory observation. ALPHA reported that antihydrogen atoms released from magnetic confinement in ALPHA-g behaved consistently with gravitational attraction to Earth, while repulsive antigravity was ruled out in that tested case [12]. The result should be read neither too weakly nor too strongly. It is not merely a philosophical reassurance that antimatter should gravitate. It is a direct experiment with neutral anti-atoms. But it is also not yet a high-precision equality test between hydrogen and antihydrogen gravitational acceleration. Its importance lies in opening a precision program, not ending the subject.

BASE's 2025 single-antiproton spin result marks a different kind of movement. The headline is not only that an antiproton was measured, but that a single antiproton spin was coherently controlled in a cryogenic Penning-trap system [17]. The reported Rabi oscillations, spin-inversion probabilities above 80 percent, roughly 50 s spin coherence, and 16-times narrower linewidths point to a future in which antiproton magnetic-moment comparisons can be limited less by basic spin control and more by deeper sources of systematic uncertainty. This is a recurring pattern in the recent literature: method development is physics development because better control changes the reachable measurement.

ALPHA's recent spectroscopy and accumulation work follows the same pattern. Precision spectroscopy of the hyperfine components of the 1S-2S transition in antihydrogen improved the data-taking rate and connected antihydrogen spectroscopy to standard-model-extension constraints [9]. The 2025 accumulation result then attacked one of the practical bottlenecks directly, reporting an eight-fold increase in trapping rate and more than 15000 antihydrogen atoms accumulated in under seven hours through beryllium-assisted positron cooling [14]. The 2026 ground-state hyperfine result used roughly 24,000 anti-atoms to reach 4 ppm precision and reported a value consistent with hydrogen expectations in the relevant magnetic-field context [10]. These are different achievements, but they point in the same direction: antihydrogen measurements are becoming more repeatable, higher-statistics, and more sensitive to subtle structure.

These developments do not remove the need for caution. A result consistent with CPT in one observable is not a proof that every matter-antimatter comparison will agree at every scale. A result consistent with gravitational attraction is not the same as a complete precision test of the weak equivalence principle for antimatter. A method that accumulates more antihydrogen atoms is not itself a symmetry test, even though it enables better symmetry tests. This review therefore treats recent results as changes in measurement capacity as much as changes in measured values. The distinction matters because many future claims will depend on systematics rather than on whether antihydrogen can be made at all.

The AD/ELENA program review emphasizes this long horizon. It frames future goals through improved antimatter precision, transportable traps, gravity studies, dark-sector possibilities, and new hadron-physics directions reaching beyond the current decade [3]. This is consistent with the structure argued throughout the paper: the program advances by widening the set of controllable antimatter systems and tightening the interpretation of each observable. BASE-STEP and transportable antiproton traps, for example, matter because moving antiprotons to quieter precision laboratories could reduce environmental limitations. GBAR and AEgIS matter because independent gravity methods can check whether ALPHA-g's trajectory-based conclusions hold under different kinematic assumptions. PUMA matters because it expands the use of antiprotons into nuclear structure rather than only fundamental-symmetry tests.

The largest open question remains the same one that motivates the field: why the observable universe is matter-dominated. Current low-energy antimatter experiments do not solve that problem directly. Instead, they test pieces of the theoretical structure that any successful account must respect or revise. The near-term research program is therefore not a single search for a dramatic anomaly. It is a disciplined tightening of comparisons: proton against antiproton, hydrogen against antihydrogen, trapped release against beam deflection, spectroscopy against gravity, and nuclear annihilation against conventional nuclear probes. If a deviation appears, the value of this portfolio will be that the field has multiple ways to ask whether it is real. If no deviation appears, the value will be increasingly strict boundaries on where new physics can hide.

7. Conclusion

CERN's antimatter program is strongest when read as a portfolio of precision systems. The basic motivation is cosmic: the observed universe is matter-dominated, and the origin of that asymmetry remains one of the deepest open questions in physics. But the practical work of CERN's low-energy antimatter program is more specific. It turns broad symmetry questions into measurable comparisons involving antiprotons, antihydrogen, antiprotonic atoms, and antiproton-nucleus annihilations.

The central conclusion of this review is that complementarity is not a secondary feature of the program; it is the program's main scientific value. BASE constrains baryonic symmetry through trapped charged particles. ALPHA uses neutral antihydrogen to test spectroscopy and gravity. AEgIS and GBAR offer independent gravitational strategies with different kinematics and systematics. ASACUSA extends precision spectroscopy into antiprotonic atoms. PUMA treats antiprotons as probes of rare nuclear structure. Each experiment measures something the others cannot measure in the same way.

Recent results sharpen this picture. BASE's 16 parts-per-trillion charge-to-mass comparison and coherent single-antiproton spin spectroscopy show how single-particle control drives baryonic CPT tests. ALPHA's gravity, accumulation, and hyperfine measurements show antihydrogen becoming a more mature precision platform. PUMA and ASACUSA remind us that antimatter research is not only about antihydrogen or gravity. The frontier is not one dramatic test, but a disciplined accumulation of cross-checks.

That discipline also requires careful limits. Current measurements support matter-antimatter symmetry within the observables and uncertainties tested, and ALPHA's gravity result rules out simple repulsive antigravity for antihydrogen in the tested case. They do not prove CPT in every possible theoretical setting, measure all aspects of antimatter gravity to arbitrary precision, or solve baryogenesis directly. Their importance is more subtle and more durable: they make the space of possible new physics smaller, clearer, and more experimentally accountable.

The next stage of this research should narrow from map to analysis. A follow-up paper could focus specifically on antihydrogen gravity, proton-antiproton precision tests, or a reproducible database of antimatter measurement progress. This first review provides the foundation for that work by identifying the systems, observables, and limits that any serious antimatter project must keep separate.

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